

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I V. Sheethika / 192475056 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sheethika

Industry Problem Solving Task

1. 20-Person Startup — Star Topology

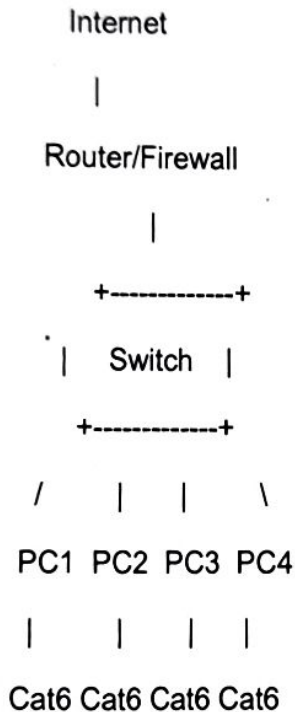
Problem:

A startup needs a simple, affordable, manageable network for around 20 employees, including computers, printers, VoIP phones and wireless devices.

Proposed

Use a star topology with a managed 24/48-port switch at the center.

Solution:



Technologies:

- Managed Layer 2/Layer 3 switch
- Cat6 UTP
- Router/firewall
- Wi-Fi access point
- DHCP
- NAT
- Gigabit Ethernet

Justification:

If one cable or workstation fails, the other devices continue operating. Centralized management makes troubleshooting and future expansion easy.

Validation:

Disconnect one workstation cable and verify that the remaining devices continue communicating.

2. Bank — Mesh Topology

Problem:

A bank requires extremely high availability because network failure can affect financial transactions and services.

Solution:**Proposed**

Use a **partial/full mesh topology** at the core with redundant routers, switches and links.

Router 1

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Core SW1 --- Core SW2

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| \ / |

| \ / |

Firewall Firewall

Technologies:

- Fiber optic cables
- Redundant routers
- Redundant core switches
- OSPF/BGP
- Firewall
- UPS
- RAID
- Single-mode/multi-mode fiber

Justification:

If one link fails, traffic can use another available path. This provides high availability and minimizes downtime.

Validation:

Disconnect one core link and verify that traffic continues through the alternate route.

3. Classroom Lab — Bus Topology

Problem:

A small classroom requires a low-cost network with limited hardware requirements.

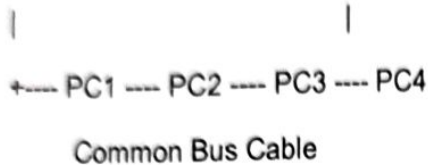
Proposed

A traditional **bus topology** can be used where appropriate, using a common backbone cable.

Solution:

Terminator

Terminator



Technologies:

- Coaxial cable
- BNC connectors
- T-connectors
- 50-ohm terminators

Advantages:

- Low hardware cost
- Simple design
- Suitable for small/light workloads

Limitations:

- Backbone failure affects the network
- Collisions can occur
- Difficult to troubleshoot as the network grows

Validation:

Test communication between multiple computers and observe the effect of a backbone cable failure.

Note: For a modern classroom network, a switched **star topology** using Cat6 and an unmanaged switch is generally more practical than a traditional bus network.

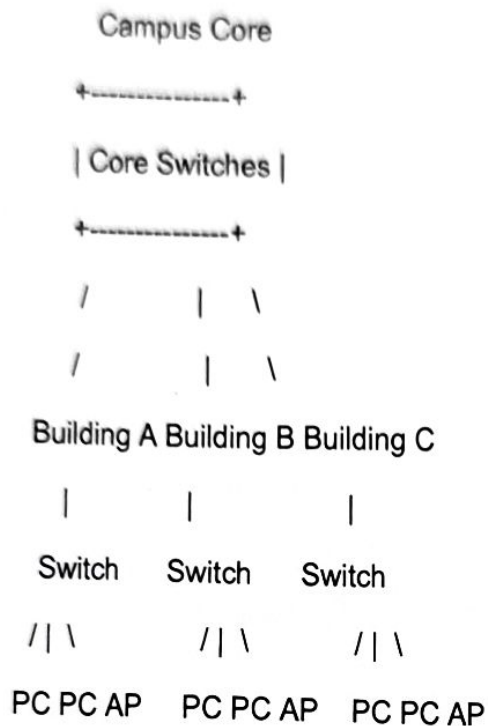
4. University Campus — Hybrid Topology

Problem:

A university has thousands of users distributed across many buildings, so one topology cannot efficiently serve the entire campus.

Proposed

Use a **hybrid topology** combining mesh/ring at the backbone and star topology within buildings.

Solution:**Technologies:**

- Fiber optic backbone
- 10/40 Gbps links
- Layer 3 switches
- Cat6
- Wi-Fi access points
- PoE
- VLANs
- Core/distribution/access architecture

Justification:

The backbone provides redundancy while star topology inside buildings makes management easy. New buildings can be added without redesigning the complete network.

Validation:

Disconnect one fiber uplink and verify that the building remains reachable through its redundant path.

5. Company with Multiple Departments — VLAN-Based Hierarchical Network

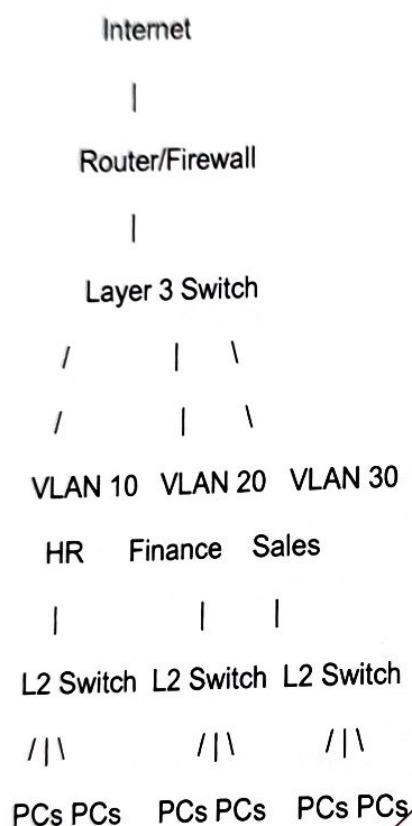
Problem:

HR, Finance, Sales and IT need to communicate while keeping departmental traffic logically separated and secure.

Solution:

Proposed

Create separate **VLANs** for each department and use a Layer 3 switch for inter-VLAN routing.



Example VLAN Configuration

Department	VLAN	Example Network
HR	10	192.168.10.0/24
Finance	20	192.168.20.0/24

Sales	30	192.168.30.0/24
IT	40	192.168.40.0/24

Technologies:

- Layer 2 switches
- Layer 3 switch
- VLANs
- Inter-VLAN routing
- Router/firewall
- DHCP
- Subnetting
- Redundant links

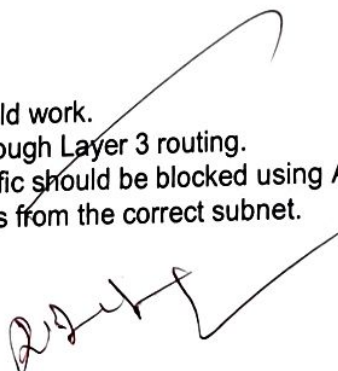
Justification:

VLANs isolate departments, reduce unnecessary broadcast traffic and improve security. Inter-VLAN routing allows authorized communication between departments.

Validation:

Test:

1. HR → HR communication should work.
2. HR → Finance should pass through Layer 3 routing.
3. Unauthorized departmental traffic should be blocked using ACL/firewall rules.
4. Verify DHCP assigns addresses from the correct subnet.



RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution